

Networks and Security

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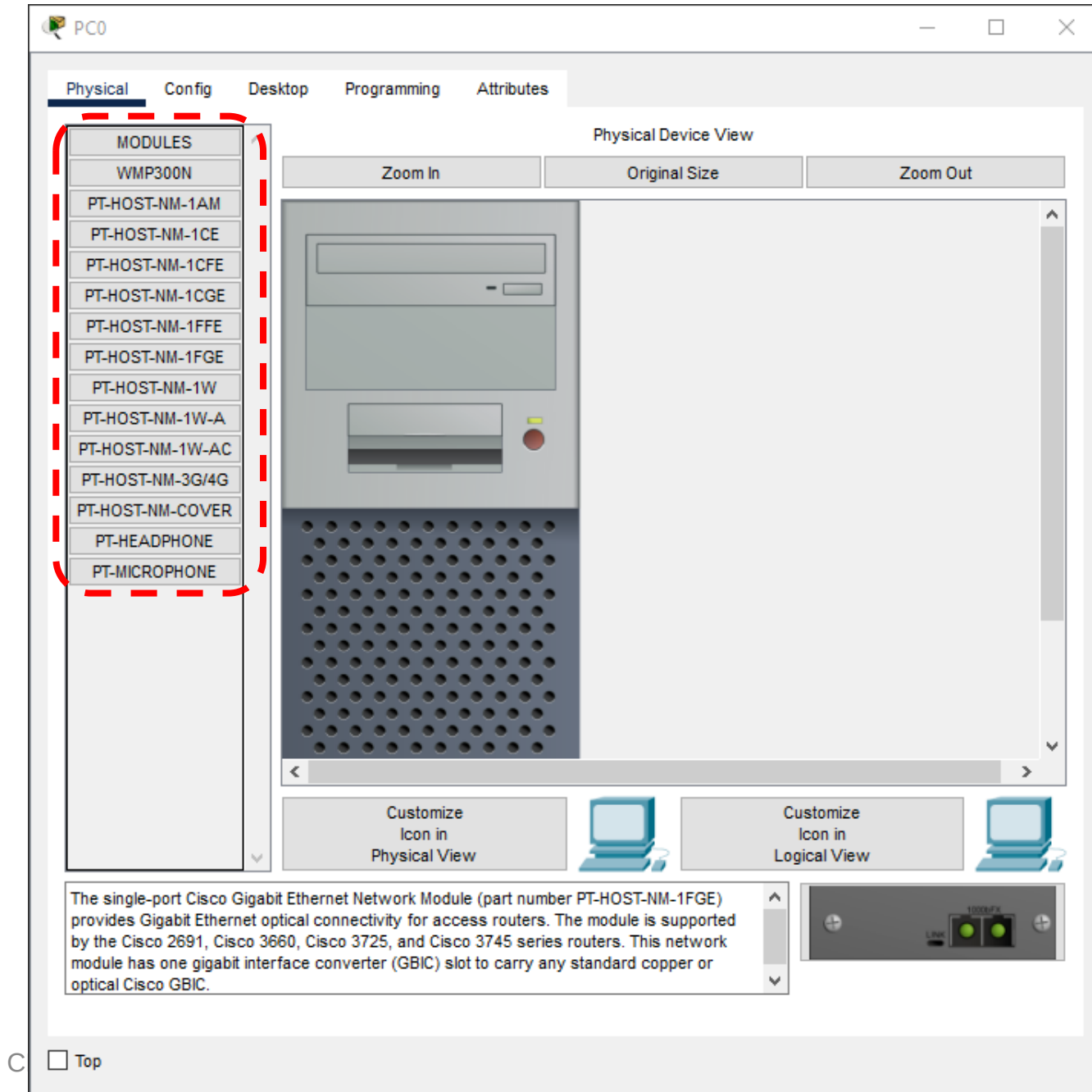
The *Career* University

Lab#2

Generic IP End Devices

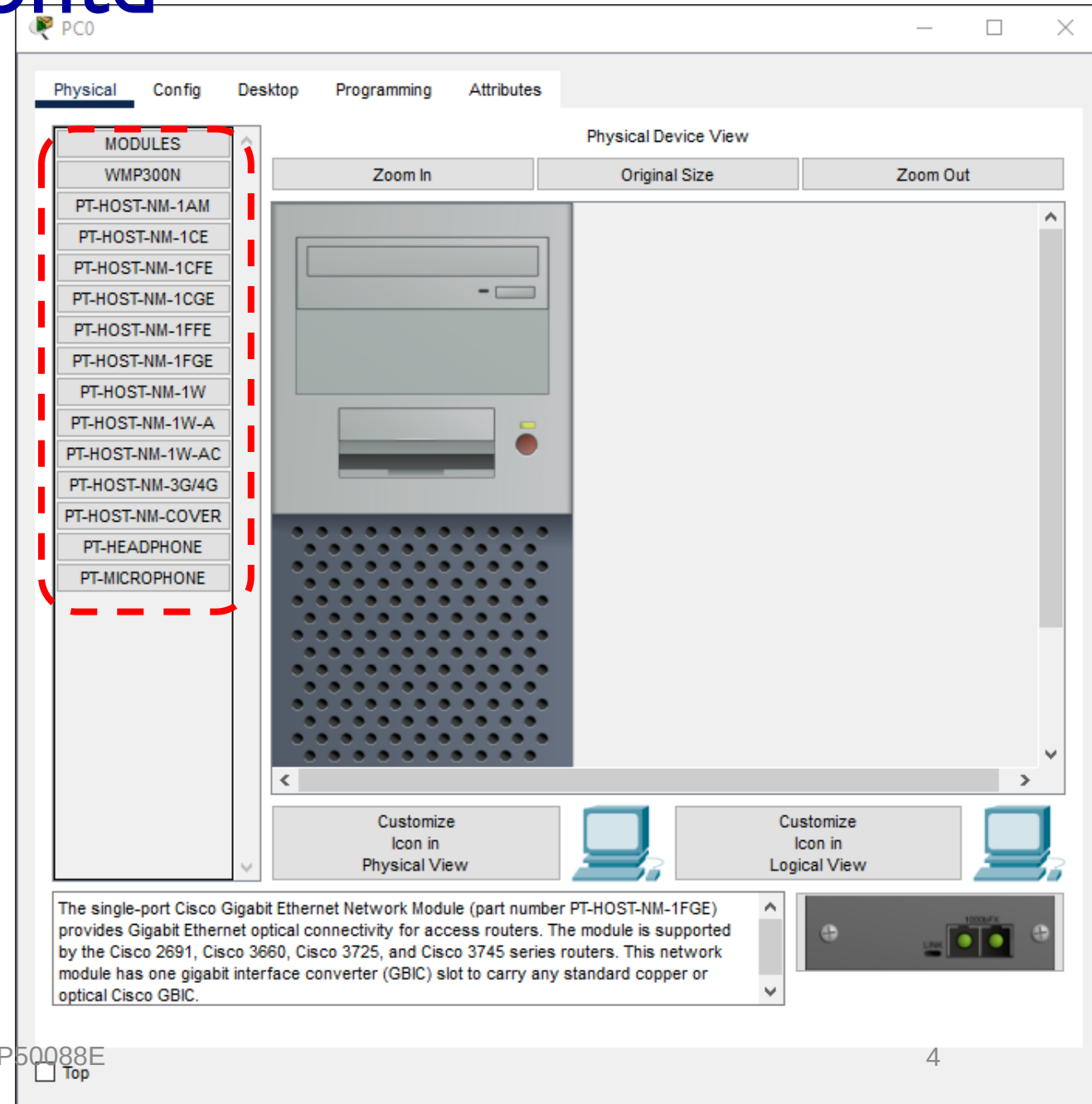
Desktops and laptops

- **Linksys-WMP300N:** This is the normal LAN interface. Based on speeds, these interfaces are named Ethernet (10 MBps), FastEthernet (100 MBps), and GigabitEthernet (1000 MBps).
- **PC-HOST-NM-1AM:** This provides an RJ11 interface that can be used as a dial-up modem.
- **PC-HOST-NM-1CE, PC-HOST-NM-1CFE, PC-HOST-NM-1CGE:** These three modules provide an **Ethernet**, **FastEthernet**, and **GigabitEthernet** connection, respectively.



Desktops and laptops-contd'

- **PC-HOST-NM-1FFE, PC-HOST-NM-1FGE:** This is the fibre Ethernet version of the previous module.
- **PC-HOST-NM-1W, PC-HOST-NM-1W-A:** Both of these modules provide a wireless interface for WLAN. The first one has a frequency of 2.4 GHz and the second 5GHz for 802.11a networks.



Connecting devices

- **Console:** This is a console cable that is used to view the network device's console from a PC/laptop. One end of the cable connects to the console port of a network device while the other one connects to the RS-232 port on a PC/laptop.
- **Copper straight-through:** This is a standard Ethernet cable that is used to connect two devices that operate in different layers of the OSI model (such as hub to router and switch to PC). It can be used with Ethernet, Fast Ethernet and Gigabit Ethernet port types.



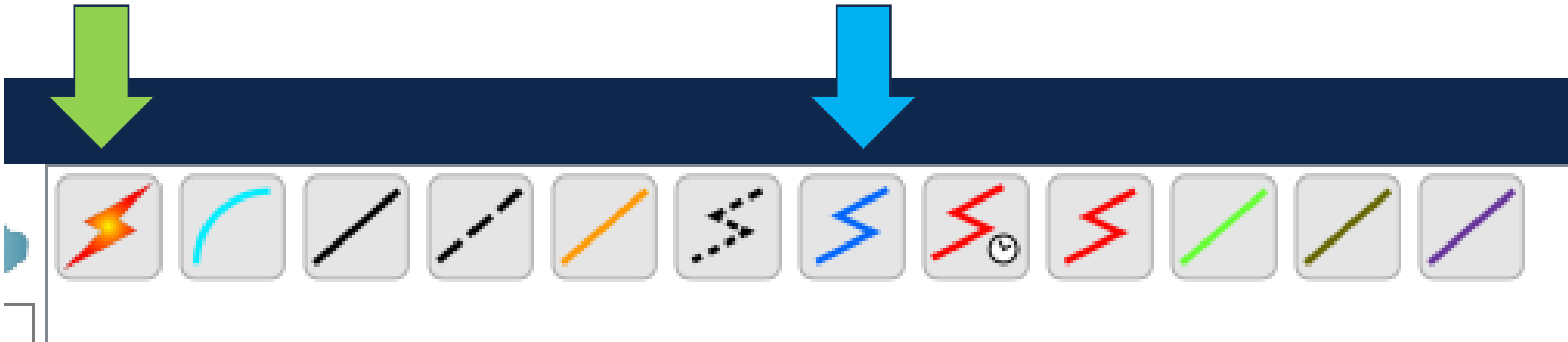
Connecting devices- contd'

- **Copper cross-over:** This Ethernet cable connects devices operating in the same OSI layer (such as hub to hub, PC to PC, PC to router, and PC to printer). This cable can also be used with Ethernet, Fast Ethernet and Gigabit Ethernet port types.
- **Fibre:** This cable connects Fast Ethernet and Gigabit Ethernet ports of a fibre port.



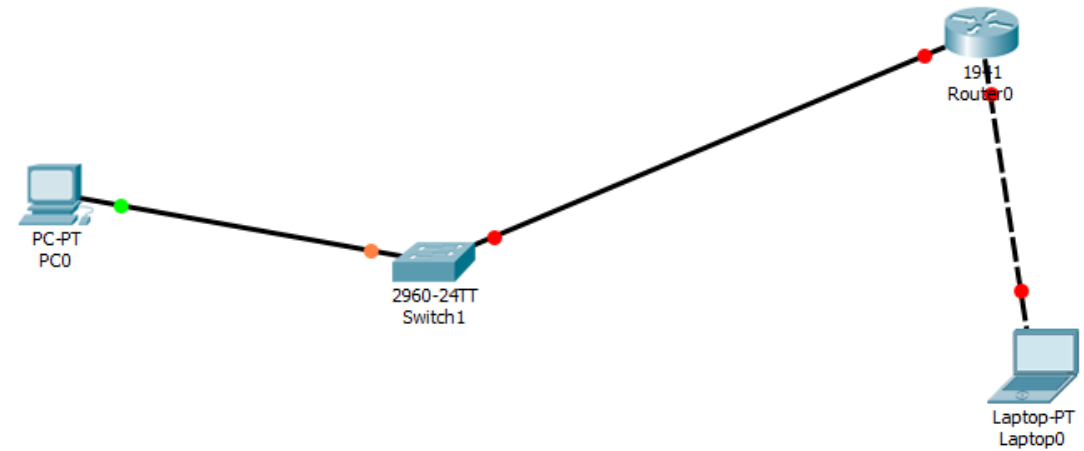
Connecting devices- contd'

- **Coaxial:** The coaxial cable connects the cloud with a cable modem and a TV with the cloud.
- **Automatically choose connection type:** If you are confused about the cable to use, choosing this option automatically connects two devices with the best cable.



Link status

- **Bright green:** This indicates that the physical link is up, but it doesn't indicate the status of the line protocol.
- **Blinking green:** This indicates link activity.
- **Red:** This indicates that the physical link is down. This can be caused by incorrect cables or by a port being administratively shut down.
- **Amber:** This appears only on switches, and indicates that the port is running the STP (Spanning Tree Protocol) algorithm to detect layer 2 loops.



Testing connectivity with protocol data unit (PDU)

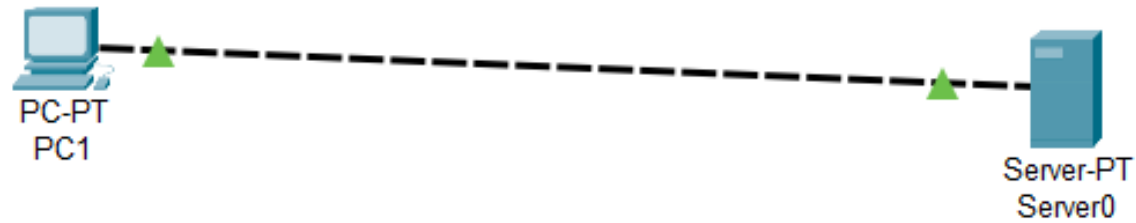
- Once a topology has been created, connectivity can be tested between devices by using either simple or complex PDUs.
- Although it is possible to do the same by pinging devices from their command-line interface, using the PDU option is quicker for large topologies.

Simple PDU

We will create a topology with a PC and a server to demonstrate how this option works:

1. Add a PC and a server to the workspace and connect them using a copper crossover cable.
2. Assign IP addresses to both of them. Example, PC1:192.168.0.1 and Server: 192.168.0.2.
3. From the common tools bar, click on the closed envelope icon or use the shortcut key P.
4. The pointer will change to an envelope symbol. Click on the PC first and then on the server. Now look at the User Created Packet box. You'll see the status as Successful and will also see the source, the destination, and the type of packet that was sent.

Topology- Simple PDU



Realtime Simulation

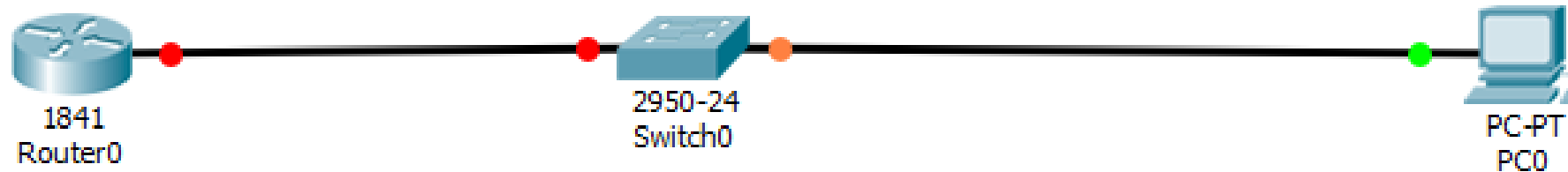
Scenario 0

New Delete

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	Server0	ICMP		0.000	N	0	(edit)	(delete)

Connection between Router, Switch & Device

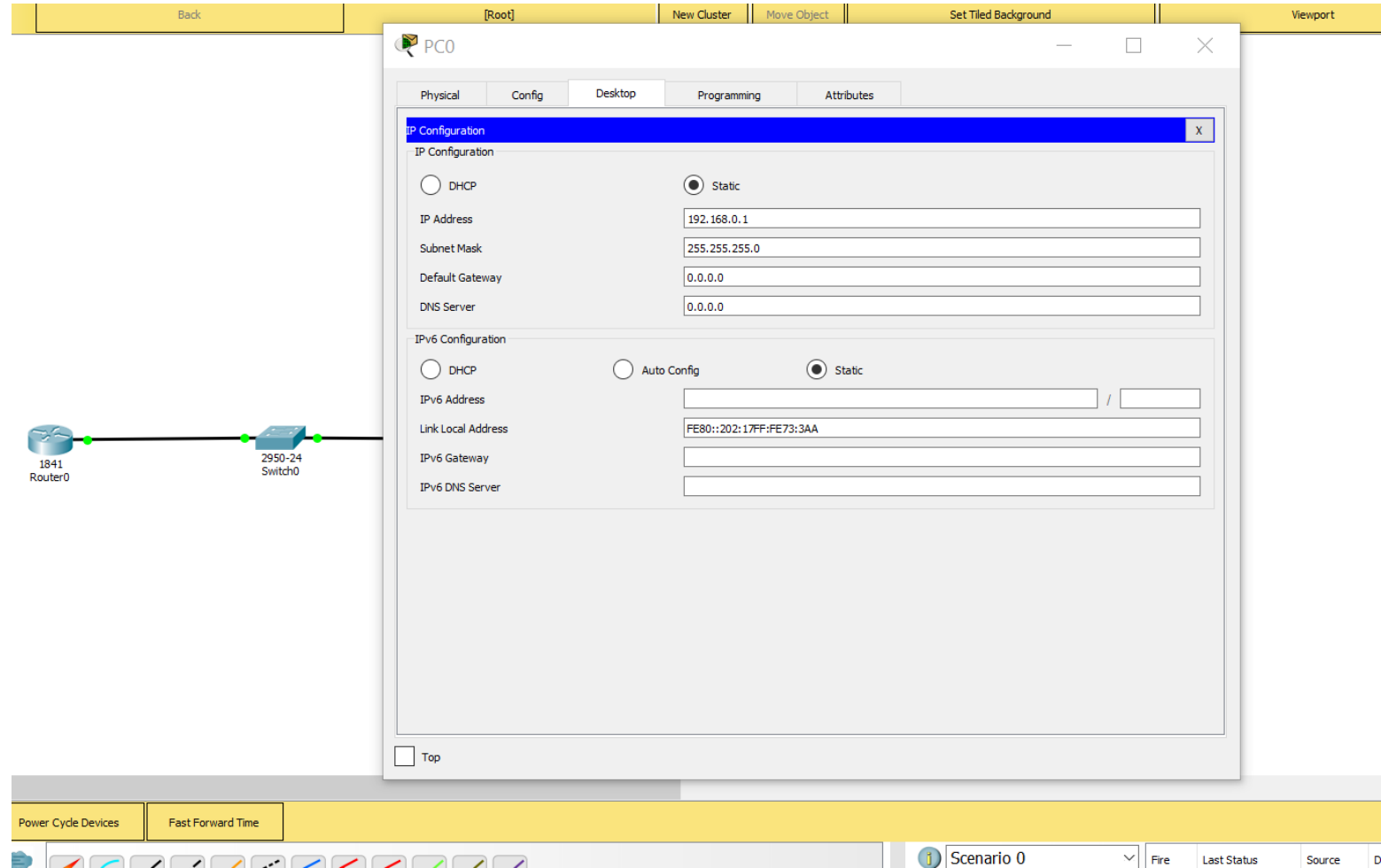
Now, we will create a topology with a PC and a server to demonstrate how this option works:



Connection between Router, Switch & Device

Step 1:

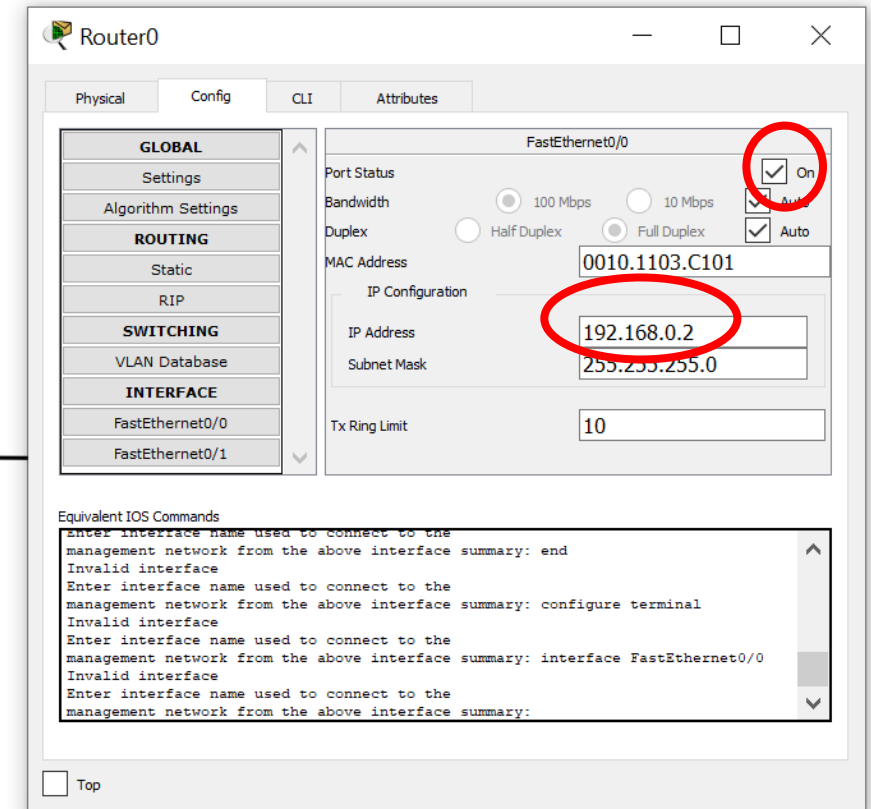
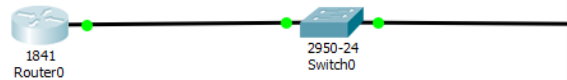
- Open the PC/laptop, navigate to the *IP configuration* tab, and then assign an IP (e.g., 192.168.0.1)



Connection between Router, Switch & Device

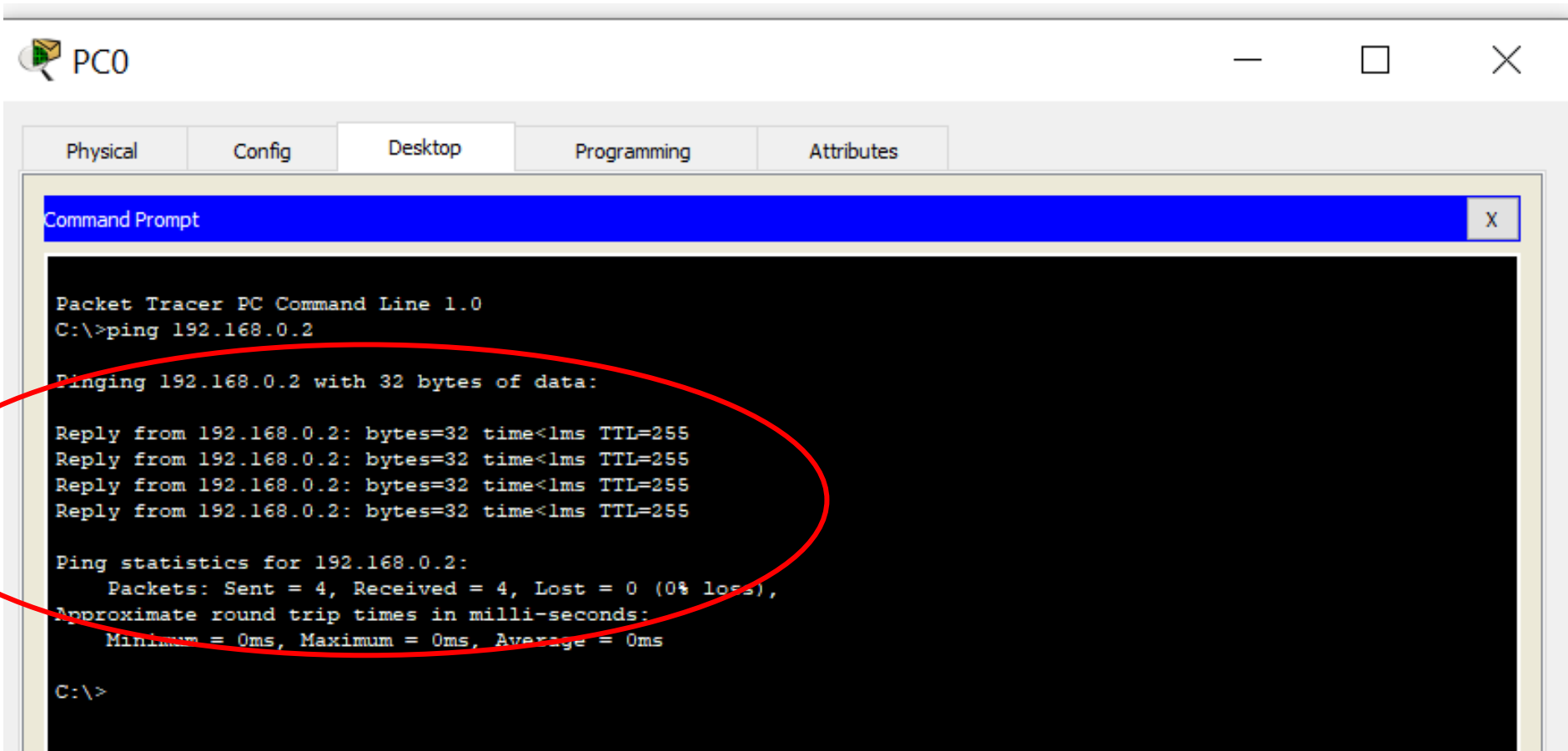
Step 2:

- Open the Router, navigate to the *Config* tab, and then click on FastEthernet0/0, then assign an IP (e.g., 192.168.0.2)



Connection between Router, Switch & Device

Last step: Check the connection



The screenshot shows a Packet Tracer PC window for PC0. The 'Desktop' tab is selected, displaying a Command Prompt. The command prompt shows the execution of the command 'ping 192.168.0.2'. The output indicates that the ping was successful, with four replies received from 192.168.0.2, each with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The ping statistics show that all four packets were sent and received, with 0% loss. The approximate round trip times in milliseconds are all 0ms. A red oval is drawn around the ping output text.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=255
Reply from 192.168.0.2: bytes=32 time<1ms TTL=255
Reply from 192.168.0.2: bytes=32 time<1ms TTL=255
Reply from 192.168.0.2: bytes=32 time<1ms TTL=255

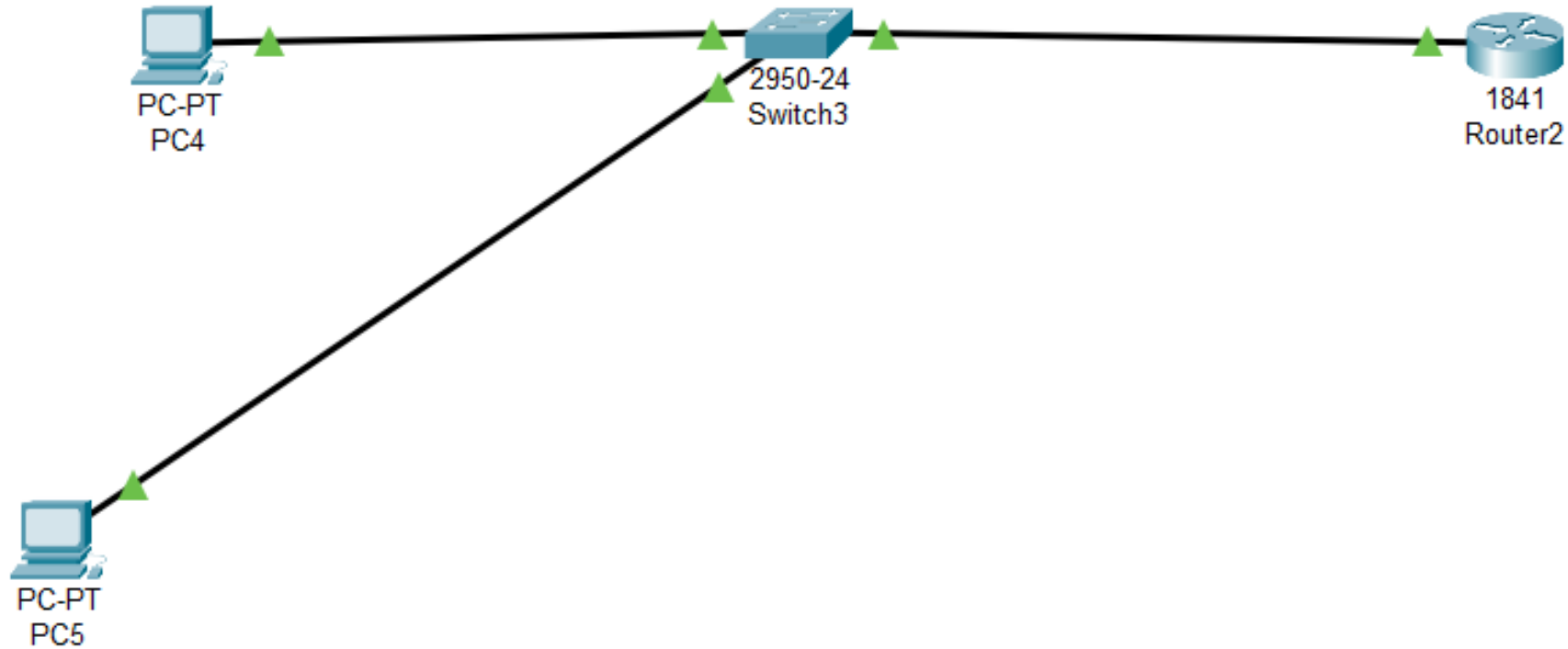
Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Practical Time

Practice

Try to connect the two PCs to a router through a switch.



Your Task

After completing the creation of the topology and IP assignments, **show your tutor** that both PCs can be pinged from the router and vice versa to **earn 1 mark**.

Reference

- **Computer Networking: A Top-Down Approach**

8th edition

Jim Kurose, Keith Ross

Pearson, 2020

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